Shaokang Jiang

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Education

University of California, Irvine — Irvine, CA
Ph.D. in Software Engineering, Advised by Daye Nam — *2025-2030 (expected)*

University of California San Diego — La Jolla, CA
M.S. in Computer Science and Engineering, Advised by Michael Coblenz — *2022-2025*
Thesis: Usability Analysis of Programming Assistants

University of Wisconsin-Madison — Madison, WI
B.S. in Computer Science, GPA: 4.00/4.00, graduated with distinction in the major — *Graduated 2021*

Research Experience

Graduate Student Researcher — August 2025 – now
University of California, Irvine (supervised by Daye Nam)

- Analyzed human-AI collaboration using Cursor rules as a proxy, identifying four primary categories (Project, Convention, Guideline, LLM Directive) and revealing that developers adapt context types by programming language while applying traditional human-to-human documentation strategies to instruct AI. Published at **MSR 2026**.
- Designed and implemented a pipeline to collect and analyze Cursor rules from 401 GitHub repositories, leveraging prompted LLMs to scale up the qualitative coding process.
- Developed an ambient automation discovery system utilizing LLMs to passively observe web behavior and generate personalized programs, validated via an 8-person empirical study, revealing users inherently fail to recognize over 94% of their own automatable actions. Submitted to **UIST 2026**.
- Designed and implemented an asynchronous, natively cross-platform multimodal discovery agent (Rust, JavaScript, Chrome APIs) utilizing contextual memory and different LLM reasoning (e.g.,chain-of-thought, self-reflection, programming-as-reasoning, planning, and cross-verification) to mine web interactions into patterns. The system designs and generates scripts following human feedback, showing a 93% usefulness rating for all viewed patterns.

Graduate Student Researcher — April 2024 – May 2025
University of California San Diego (supervised by Michael Coblenz)

- Found Copilot helps users focus on learning while increasing productivity; revealed developers prefer analyzing generated code over explanations, yet still rely on search engines for conceptual questions. Submitted to **TOSEM**.
- Designed an eye-tracking experiment to evaluate the costs and benefits of Copilot for novice programmers, developing a universal eye-tracking platform for native VSCode to capture user interaction data across in-editor events.

Graduate Student Researcher — April 2024 – May 2025
University of California San Diego (supervised by Kristen Vaccaro and Deepak Kumar)

- Audited YouTube's recommendation strategies for short and long-form content; discovered that long-form algorithms reinforce partisan echo chambers by aligning with seed videos, while short-form algorithms prioritize engagement by exposing users to more right-leaning content. Published at **WWW 2026**.
- Designed and implemented a pipeline of seeds collecting, matching, recommendation collecting, and political assessing, leveraging Puppeteer to harvest recommendations, git-diff for video matching, and prompt-engineered GPTs and BERT for political labeling.

Researcher — April 2024 – December 2024
Independent (supervised by Jimmy Koppel and Michael Coblenz)

- Evaluated the impact of AI assistance on code comprehension among 15 experienced engineers, finding that LLM chatbots increased user confusion, while traditional tutorials improved high-level structural understanding but lacked low-level coding support. Published at **Plateau 2025**.

Graduate Student Researcher — April 2024 – October 2024
University of California San Diego (supervised by Steven Dow)

- Found LLM-driven conversational cues significantly increased the topical diversity of generated ideas and conversations. Published at **CI 2025**.
- Designed and deployed a meeting platform with prompt-engineered GPT conversational cues, Firebase and Daily.js.

- Led quantitative analysis evaluating the impact of AI-generated cues in a 172-participant 2x2 experiment, authoring the design, implementation, and data analysis sections for publication.

Graduate Student Researcher January 2023 – September 2023

University of California San Diego (supervised by Michael Coblenz)

- Found that autocomplete speeds up information acquisition but may not increase productivity with 32 programmers working with an unfamiliar API. Published and presented the paper at **FSE 2024**.
- Found that participants who used autocomplete learned more while spending less time reading documentation.
- Developed a web-based VSCode interaction and eye-tracking platform (JS/C#, Tobii Eye Tracker 5) to capture in-editor events, achieving a 90% latency reduction than previous solutions.

Publications

[1] S. Jiang and D. Nam, "Motif: Discovering and automating personal web workflows," *arXiv preprint arXiv:2607.10531*, Jul. 2026, Preprint.

[2] S. Jiang and D. Nam, "Beyond the prompt: An empirical study of cursor rules," in *23rd International Conference on Mining Software Repositories (MSR 2026)*, 27% acceptance rate, 2026.

[3] S. Jiang and M. Coblenz, "How llms affect developers' productivity and learning," *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026, Under review.

[4] S. Jiang, A. Arya, S. Kweon, I. Liang, D. Kumar, and K. Vaccaro, "Long story short: Auditing U.S. political polarization in recommendations for long- vs. short-form videos on youtube," in *Proceedings of the ACM Web Conference (WWW 2026)*, 20% acceptance rate, 2026, pp. 6147–6158.

[5] J. A. Rayan, S. Jiang, N. Balaji, J. Wang, I. Gross, C. Biehle, and S. P. Dow, "Cueing the crowd: Llm-driven conversational cues across different meeting modalities increase topical diversity of generated ideas," in *Proceedings of the ACM Collective Intelligence Conference (CI 2025)*, 2025, pp. 297–316.

[6] S. Jiang, J. Koppel, and M. Coblenz, "Questions before answers? a comparison of codebase chatbots and tutorials for codebase learning," in *Plateau Workshop (Plateau 2025)*, 2025.

[7] S. Jiang and M. Coblenz, "An analysis of the costs and benefits of autocomplete in ides," in *Proceedings of the ACM on Software Engineering (FSE 2024)*, 25% acceptance rate, vol. 1, 2024, pp. 1284–1306.

Other Experience

Teaching Assistant September 2025 – now

University of California, Irvine

- TA for programming in Java (ICS 45J) with Emily Navarro.
- TA for information visualization (INF 143) with Alberto Krone-Martins.
- Led discussions, graded assignments, and co-designed course content and logistics with faculty.

Teaching Assistant January 2023 – March 2025

University of California San Diego

- Major TA for Java and Data Structure courses (DSC30) with Soohyun Liao for two quarters.
- Sole TA for the Antisocial Computing (CSE 291) with Kristen Vaccaro for one quarter.
- Sole TA for the Usability of Programming Languages course (CSE291) with Michael Coblenz for one quarter and TA for the same course for another quarter.
- Sole TA for Compilers (CSE 131) with Yufei Ding for one quarter, and Introduction to Data Visualization (DSC 106) with Sam Lau for another quarter.
- Designed and implemented course assignments and mentored students through group projects.
- Facilitated weekly discussion sections and office hours, collaborating with the lead instructor to refine course content, manage logistics, and evaluate student performance.
- Received 'Excellent' ratings from instructors and a 90%+ student recommendation rate in course evaluations.

Technical Skills

- **Languages**: Java, Python, Rust, C, C++, C#, HTML/CSS/JavaScript, TypeScript, Matlab, LaTeX, GAMS, Markdown, Haskell, SQL, Embedded JavaScript, Blogger Template Language, Shell
- **Frameworks**: React, Node.js, Vue, Jest, Vitest, JSP, Flask, Expo (React Native), JUnit, Svelte, VitePress, Hexo, Electron, Chrome Extensions, VSCode Extensions, Puppeteer, Selenium, Tampermonkey, JavaFX, Servlets
- **Platforms/Tools**: Autogen, vllm, ollama, Git, Docker, GitHub, Google Cloud, Azure, Cloudflare (Workers and KV), Firebase, Oracle Cloud, VS Code, Visual Studio, IntelliJ, Eclipse, Nginx
- **Libraries/Services**: Auth0, Stripe, Matomo, MailJet, Daily.js, Socket.io, TensorFlow, WebGL, NLP.js, echarts, Matplotlib, echarts, PyQt, Pyramid, CherryPy, Three.js, and much more
- **Other Platforms**: Windows, Linux, MAC, JIRA, Trello, DevOps, Agile, JMP, Figma, Photoshop, Qualcoder, Taguette, Premiere, Audition, Blender, Excel, Word, PowerPoint

Projects

Highlighted below are selected projects. For a comprehensive portfolio, visit shaokangjiang.github.io/resume/.

Monitoring Extension | *JavaScript, Rust, Chrome Extension, Apps Script, Express.js, Multimodal LLM* 2026

- Designed and developed a Chrome extension to monitor real-time user interactions, utilizing a JavaScript frontend bundled with rollup, a Rust-based multi-threads native host for cross-platform communication and high-fidelity image/data logging, and an Express.js backend bundled with SEA for asynchronous, streaming data processing.
- Designed and implemented a privacy-first LLM automation pipeline using Gemini to passively detect and automate recurring user patterns through action abstraction, pattern mining, and program generation. Developed a stateless architecture utilizing local-only contextual memory and user-controlled telemetry to ensure full data sovereignty.
- Optimized multimodal AI performance with reasoning (chain-of-thought, self-reflection, programming-as-reasoning, planning, and cross-verification) to analyze inputs from network events, user interactions, and screenshots.

Youtube Recommendation Analysis | *JavaScript, Selenium, Node.js, Torch, Transformers* 2025

- Developed a data pipeline to sample, pair, and harvest 50K YouTube recommendations using sock-puppet accounts.
- Developed a git-diff-inspired text algorithm to identify matching video pairs, achieving 97% accuracy (0.95 precision, 0.96 recall), and trained a custom BERT model to classify the political leaning of content.

Workshop Manager | *Electron, React, Socket.io, HTTPS, DDNS, GPTs, Worker, MailJet* 2024

- Designed and developed a 4-component workshop management system for real-time user engagement to foster idea generation and improve communication during meetings, capable of hosting at least 50 concurrent participants.
- Pre-conference component: Created a registration system utilizing a RESTful API backend with Worker for user verification and sending invitation emails via Mailjet. The frontend was developed using a simple HTML page.
- In-conference Management Component: Uses DDNS to dynamically assign a subdomain. Developed with Electron as the main platform, with React for the frontend and a JavaScript backend running an HTTPS WebSocket server. The management system will communicate with and broadcast guidance to all participants with the help of GPT services for hint generation, while also updating the main screen with large-screen guidance.
- In-conference Client-side Ideation and Discussion Components: Both components allow participants to submit ideas in real-time during small group discussions and facilitate large group discussions on the submitted ideas in a subsequent phase. They were developed using React and Socket Client for real-time communication.

HealthCare Chatbot | *Expo, Worker, GitHub, Jest, Vitest, Figma* 2024

- Led a team of five to create a chatbot with CI/CD pipelines using Agile methodologies. Self-delegate leadership roles when necessary. Consulted with experts in health care to understand their needs and requirements.
- Developed a chatbot using Expo to provide healthcare information and answer questions via a fine-turned GPT service. Backend services were built with JavaScript and deployed on Cloudflare Workers to handle user information management, session storage, authentication and token management, using salted hashes for password storage.
- Learned methods for managing disputes and mediation conflicts within a team, including documenting divergence points and finding common ground. Realized the importance of communication in a team.
- Integrated Material Design principles and conducted unit tests using Jest for the Expo app and Vitest for the middleware server. Utilized GitHub Actions for the CI/CD pipeline.

Online interactive shell | *Java Server Page, Servlets, Java, Azure* 2023

- Developed an interactive shell using JSP and Servlets deployed on Azure, enabling students to run Java source code online without accessing the source code, while supporting over 30 users simultaneously. The backend featured a custom connection session management utility to optimize resource usage and ensure seamless session handling.

An economical computer | *Node.js, GAMS, Puppeteer* 2022

- Automatically scraped computer components' performance data weekly using Puppeteer, with GitHub Actions and a cron scheduler, by simulating user behavior.
- Used optimization with preset LP problem-solving to find the best configuration.

Wise grader | *JavaScript, TensorFlow, Tampermonkey* 2021

- Developed and deployed a Tampermonkey script to expedite grading by suggesting scores based on length, N-gram analysis, and topic relevance for dictation questions, using the TensorFlow. The entire system operates locally.

Financial Tutoring website and app | *HTML, Ejs, Worker, JavaScript, echarts, Expo, NLP* 2020

- Designed and led a team to develop a static financial education website for Capital One. The site includes interactive learning tools, such as quizzes and a self-built, stateless learning progress tracking system. It also has various utilities, including a personal expense management app.
- Created detailed implementation guidance for group members and a user manual for the website and learning content. Build CI/CD pipelines and deployed them with GitHub Actions.
- Built front end using embedded JavaScript for the main site and HTML with JavaScript for utility tools. Developed backend with JavaScript and deployed on Cloudflare Workers to handle requests, including the encryption and decryption of learning stage information using AES and crypto and a dynamically rendered quiz system. Users can download their encrypted learning progress and upload it to decode and resume learning, or use the same browser to continue learning. No data is stored on the server considering privacy and budget.
- Developed a cross-platform personal finance app for Capital One that enables users to manage expenses and income via voice and chat interfaces using a local NLU engine, built with Expo for Android and iOS, tested and packaged for both platforms; Designed with JSON import/export for secure, local-only data storage and syncing.
- Followed Agile methodologies, discussed with mentors and communicated with stakeholders to understand their needs and software development stages.

Energy Simulator | *HTML, WebGL, JS, Java, JavaFX, Three.js, echarts* 2020

- Designed and developed a web-based energy simulation tool utilizing jsLPSolver to optimize energy sources based on historical data and project future trends via linear regression.
- Presented results on a static website built with HTML and JavaScript, featuring a 3D energy system simulation using A-Frame and Three.js, with AR functionalities via QR codes, dynamic data visualization, and lightweight localization. Ensured full support for modern browsers and partial support for IE 11.
- Reimplemented most functionalities in JavaFX without 3D representation.

AutoCommenter | *Java, JFrame* 2019

- Designed and developed a local utility program in Java with JFrame to analyze Java code on preset rules and semi-automatically add comments by integrating user input through typing or voice, with voice recognition powered by the iFlyTek API and translation supported by the Baidu Translation API.

HateFate Website | *HTML, Java Server Page, SQL, Java, Servlets* 2017

- Led a four-person team in developing a social networking website that connects users based on shared dislikes, with features including login, email-verified signup, and user pairing. Utilized Material design to create the interface.
- Developed the frontend using HTML and JavaServer Pages (JSP) and built the backend with JSP and SQL for data storage. Deployed on the Azure cloud with an Azure SQL database.
- Developed multiple login methods and utilized cookies to store user information, enabling automatic session recovery. Designed and integrated a custom mail system to handle email verification, password resets, and new match notifications using SMTP.
- Secured the website with SSL encryption and employed a combination of MD5, public/private key encryption, and AES for data security during page forwarding and email verification.

Awards and Activities

- Student volunteer at FSE 2024 July 2024
- Graduated from UW-Madison with distinction in the Major Spring 2021
- Dean's List of College of Letters & Science at UW-Madison for each quarter